Chapter 5

Sobolev Spaces

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 5; printed pp. 239–292; PDF pages 256–309.

Chinese reference: 中译本：书内第 169–207 页；PDF 第 187–225 页。

Editorial study edition. This is an original expanded exposition following Evans, Chapter 5, rather than a verbatim transcription. Source section and theorem identifiers are given in headings. Proofs, worked solutions, and Chinese annotations are editorial reconstructions. Our theorem numbers run throughout the chapter and differ from the source. The 21 source problems are concisely restated with editorial solutions in §5.10.

The model is the Dirichlet problem $-\Delta u = f$. Its weak formulation is

$$\int_U Du \cdot D\phi \, dx = \langle f, \phi \rangle, \qquad \phi \in H_0^1(U).$$

Only one derivative of u is required. This chapter makes each symbol precise and explains which estimates survive limits of approximate solutions.

先看整条逻辑链

弱导数容纳能量有限但不光滑的函数；逼近让经典计算可用；延拓处理边界；迹解释边界值；嵌入与紧性负责估计和取极限。

Reading route and prerequisites. Read §5.2–§5.7 in order; use §5.8 as a toolbox and §5.9 for evolution equations. Unless stated otherwise, $U \subset \mathbb{R}^n$ is open, derivatives are weak, and functions equal almost everywhere are identified. The notation $V \Subset U$ means $\overline{V}$ is compact and contained in U. Constants C are independent of the functions being estimated. Basic integration theory, Hölder and Minkowski inequalities, completeness of L^p, smooth partitions of unity, and explicitly named functional analysis theorems are prerequisites; they are not proved anew here.

5.1 Hölder Spaces

5.1.1 Definitions and a complete function space

Definition 5.1 (Evans §5.1)
For $0 < \gamma \le 1$, define

$$[u]_{C^{0,\gamma}(\overline{U})} = \sup_{x \neq y \in U} \frac{|u(x) - u(y)|}{|x - y|^{\gamma}}.$$

The space $C^{k,\gamma}(\overline{U})$ consists of C^k functions on U whose derivatives through order k extend continuously to $\overline{U}$ and have finite norm

$$\|u\|_{C^{k,\gamma}} = \sum_{|\alpha| \le k} \|D^{\alpha} u\|_{\infty} + \sum_{|\alpha| = k} [D^{\alpha} u]_{C^{0,\gamma}}.$$

Here $C^{0,1}$ means Lipschitz continuous.

For $0 < \gamma < 1$, $|x|^{\gamma}$ on $[-1, 1]$ is γ-Hölder but not Lipschitz at zero. Concavity gives $|a^{\gamma} - b^{\gamma}| \le |a - b|^{\gamma}$ for $a, b \ge 0$, whereas $|h|^{\gamma}/|h| \to \infty$. This distinguishes a quantitative modulus of continuity from differentiability.

半范数为何不够

常数函数的 Hölder 半范数为零，故必须加上 $\|u\|_{\infty}$ 才成为范数。γ 控制的是振荡速度，不是函数大小。

Theorem 5.2 (Evans §5.1, Theorem 1)
$C^{k,\gamma}(\overline{U})$ is a Banach space.

Expanded proof (editorial; also source Problem 1).

Let (u_j) be Cauchy. Each $D^{\alpha} u_j$, $|\alpha| \le k$, converges uniformly on $\overline{U}$ to a continuous bounded function g_{α}. Put $u = g_0$. We must verify that these limits are its derivatives.

Fix a closed ball contained in U. For a coordinate segment in the ball,

$$D^{\alpha} u_j(x + h e_i) - D^{\alpha} u_j(x) = \int_0^h D^{\alpha + e_i} u_j(x + t e_i)\, dt \quad (|\alpha| < k).$$

Uniform convergence allows passage through the finite integral. The fundamental theorem of calculus yields $\partial_i g_{\alpha} = g_{\alpha + e_i}$. Induction in the derivative order proves $u \in C^k(U)$ with the required extensions.

Given $\varepsilon > 0$, choose N with $\|u_j - u_l\|_{C^{k,\gamma}} \le \varepsilon$ for $j, l \ge N$. Pass $l \to \infty$ in the difference quotients for each fixed pair x, y, then take suprema. Lower semicontinuity of the finite sum of norm components gives $\|u_j - u\|_{C^{k,\gamma}} \le \varepsilon$. Thus u belongs to the space and the convergence is in norm. The norm axioms follow from the triangle inequality and the included supremum norm. □

Interpolation (source Problem 2). For $0 < \beta < \gamma < 1$, set $\theta = (\gamma - \beta)/(1 - \beta)$. For every $x \neq y$,

$$\frac{|u(x) - u(y)|}{|x - y|^{\gamma}} = \left(\frac{|u(x) - u(y)|}{|x - y|^{\beta}} \right)^{1-\theta} \left(\frac{|u(x) - u(y)|}{|x - y|} \right)^{\theta}.$$

Hence $[u]_{\gamma} \leq [u]_{\beta}^{1-\theta} [u]_{1}^{\theta}$. The two-term Hölder inequality on the supremum and seminorm components also gives $\|u\|_{C^{0,\gamma}} \leq \|u\|_{C^{0,\beta}}^{1-\theta} \|u\|_{C^{0,1}}^{\theta}$. For $\gamma = 1$ this is an identity.

导数的极限要认领

一致收敛保证候选导数连续；线段积分恒等式保证它确实是极限函数的导数。仅凭 $u_j \to u$ 一致，不能推出 $Du_j \to Du$。

5.2 Sobolev Spaces

5.2.1 Weak derivatives: transfer derivatives to test functions

Definition 5.3 (Evans §5.2.1)
For $u, v \in L^1_{\mathrm{loc}}(U)$, write $D^{\alpha} u = v$ if

$$\int_U u D^{\alpha} \phi \, dx = (-1)^{|\alpha|} \int_U v \phi \, dx \qquad (\phi \in C_c^{\infty}(U)).$$

The function v is a weak derivative. A distributional derivative always exists as a functional; it need not be represented by a locally integrable function.

为何没有边界项

试验函数的支集紧含于 U，所以在边界附近恒为零。弱导数的定义不规定 u 的边界值；边界条件要靠 $W_0^{1,p}$ 和迹来表达。

For a classical $C^{|\alpha|}$ function this identity follows by $|\alpha|$ integrations by parts, each contributing one minus sign. Thus the two notions agree whenever classical derivatives exist.

Lemma 5.4 (Evans §5.2.1: uniqueness)
A weak derivative is unique up to equality almost everywhere.

Expanded proof.

If v, w both work, $g = v - w$ satisfies $\int g\phi = 0$ for every test function. Take a nonnegative smooth kernel ρ, supported in the unit ball with integral one, and set $\rho_{\varepsilon}(x) = \varepsilon^{-n} \rho(x/\varepsilon)$. On $V \Subset U$, for sufficiently small ε, testing with $\rho_{\varepsilon}(x - \cdot)$ gives $g * \rho_{\varepsilon} = 0$. Approximate identities converge to g in $L^1(V)$, so $g = 0$ there. A countable exhaustion of U proves the assertion. The approximate-identity fact follows from

$$\|\rho_{\varepsilon} * g - g\|_{L^1(V)} \leq \int \rho(z) \|g(\cdot - \varepsilon z) - g\|_{L^1(V)} \, dz \to 0,$$

using a cutoff on a larger compact set, translation continuity in L^1, and domination by twice that cutoff's L^1 norm. □

Corners, jumps, and singularities: editorial examples. On $(-1,1)$, splitting the integral at zero gives

$$\int_{-1}^{1} |x|\phi'(x)\,dx = -\int_{-1}^{1} \operatorname{sgn}(x)\phi(x)\,dx.$$

The interface terms vanish because $|0| = 0$. A corner therefore has a bounded weak first derivative. Its second distributional derivative is $2\delta_0$, which is not a locally integrable function.

For $H = \mathbf{1}_{(0,1)}$ on $(-1,1)$, $-\int H\phi' = \phi(0)$. If some locally integrable g represented this, choose $|\phi_\varepsilon| \le 1$, supported in $(-\varepsilon,\varepsilon)$, with $\phi_\varepsilon(0) = 1$. Then $|\int g\phi_\varepsilon| \le \int_{-\varepsilon}^{\varepsilon} |g| \to 0$, a contradiction.

On $B_1 \subset \mathbb{R}^n$, $n \ge 2$, put $u(x) = |x|^{-a}$, $a > 0$. Away from zero, $|Du| = a|x|^{-a-1}$. For finite p,

$$\int_{B_1} |u|^p < \infty \iff ap < n, \qquad \int_{B_1} |Du|^p < \infty \iff (a+1)p < n.$$

Under the second condition, integration by parts outside B_ε leaves an inner boundary term at most $C\varepsilon^{n-1-a}\|\phi\|_\infty \to 0$. This identifies the weak gradient. Equality of the exponent gives the divergent integral $\int_0^1 dr/r$.

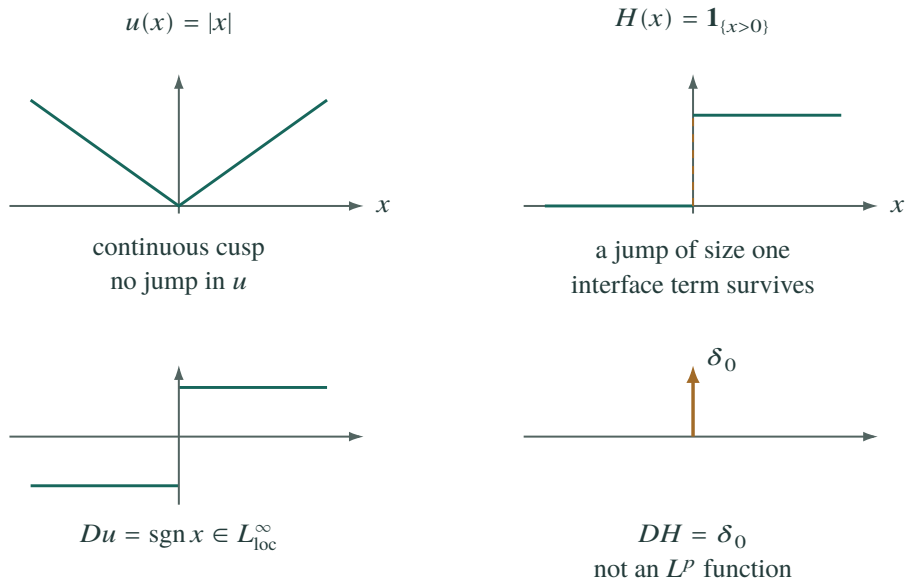


Diagram T5.1. A corner and a jump have different distributional derivatives. The continuous corner has a bounded weak first derivative. The step has a Dirac mass, so it has a distributional derivative but no weak first derivative representable by an L^p function. The vertical Dirac arrow denotes mass, not a graph of a finite-valued function.

尖角与跳跃

连续的分段光滑函数，其接口项可抵消；阶跃函数的跳跃留下 Dirac 质量。几乎处处有经典导数，不保证这个导数就是弱导数。

5.2.2 Spaces, norms, and convergence (Evans §5.2.2)

Definition 5.5

For an integer $k \geq 0$ and $1 \leq p \leq \infty$, $W^{k,p}(U)$ is the space of L^p functions whose weak derivatives through order k are all in $L^p(U)$. Use the norms

$$\|u\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_p^p \right)^{1/p} \qquad (p < \infty),$$

$$\|u\|_{W^{k,\infty}} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_\infty.$$

Write $H^k = W^{k,2}$. Define $W_0^{k,p}(U)$ as the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$. Local membership means membership on every $V \Subset U$. Functions equal almost everywhere are identified.

Our $p = \infty$ norm is equivalent to the source's convention. Strong convergence means L^p norm convergence of every derivative. For $1 < p < \infty$, weak convergence means weak L^p convergence of every component. The inner product on H^k is $\sum_{|\alpha| \leq k} \int_U D^\alpha u D^\alpha v$.

5.2.3 Calculus and completeness (Evans §5.2.3)

Theorem 5.6 (Source Theorem 1: weak calculus)

Weak derivatives are linear, restrict to open subsets, and commute: $D^\beta D^\alpha u = D^{\alpha+\beta} u$ if the total order is at most k. For $\zeta \in C_c^\infty(U)$,

$$D^\alpha(\zeta u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \zeta D^{\alpha-\beta} u.$$

Expanded proof.

Add the defining identities for linearity. A test function on an open subset extends by zero to one on U, proving locality. For commutation,

$$\int D^\alpha u D^\beta \phi = (-1)^{|\alpha|} \int u D^{\alpha+\beta} \phi = (-1)^{|\beta|} \int D^{\alpha+\beta} u \phi.$$

For the product, test with $\zeta\phi$:

$$\int \zeta u \partial_i \phi = \int u \partial_i(\zeta\phi) - \int u(\partial_i \zeta)\phi = -\int (\zeta \partial_i u + u \partial_i \zeta)\phi.$$

Iterate. The coefficients combine by the multiindex Pascal identity, giving the formula. All terms lie in L^p because all cutoff derivatives are bounded. □

零下标是闭包

先用紧支光滑逼近定义 $W_0^{1,p}$，不能提前把它解释成逐点边界值为零。有界 C^1 域、$p < \infty$ 时，迹定理才证明两种表述等价。

Theorem 5.7 (Source Theorem 2)
$W^{k,p}(U)$ is complete, including $p = 1$ and $p = \infty$.

Expanded proof.

The norm axioms follow from those of the finite product of L^p spaces. If (u_j) is Cauchy, completeness of L^p gives $D^\alpha u_j \to g_\alpha$ in L^p for every $|\alpha| \le k$. Put $u = g_0$. For $\phi \in C_c^\infty(U)$,

$$\int u D^\alpha \phi = \lim_j \int u_j D^\alpha \phi = (-1)^{|\alpha|} \lim_j \int D^\alpha u_j \phi = (-1)^{|\alpha|} \int g_\alpha \phi.$$

Hölder justifies each limit on the test function's compact support; for $p = 1$ use its L^∞ norm, and for $p = \infty$ its L^1 norm. Thus $g_\alpha = D^\alpha u$ and the finite sum of component errors tends to zero. □

For $1 < p < \infty$, the product of L^p spaces is reflexive, hence its closed Sobolev subspace is reflexive. Bounded sequences have weakly convergent subsequences. This uses reflexivity and weak compactness from functional analysis; strong convergence still requires additional work.

闭图思想

把 u 映到 $(u, Du, \dots, D^k u)$，所得像是有限个 L^p 的乘积中的闭子空间。闭性靠弱导数恒等式取极限，而不是对极限逐点求导。

5.3 Approximation

5.3.1 Interior smoothing (Evans §5.3.1, Theorem 1)

Use the kernel ρ above. If $u \in W^{k,p}(U)$, $V \Subset U$, and $\varepsilon < \operatorname{dist}(V, \partial U)$, then

$$D^\alpha(\rho_\varepsilon * u) = \rho_\varepsilon * D^\alpha u \quad \text{on } V,$$

$$\rho_\varepsilon * u \to u \quad \text{in } W^{k,p}(V) \quad (1 \le p < \infty).$$

卷积不能越界

卷积会读取半径 ε 内的数据。任意补零可能在边界制造跳跃，因此不能直接把补零后的 Du 当成补零函数的导数。

Expanded proof.

For fixed $x \in V$, differentiate the kernel under the integral; its smooth compact support and local integrability of u justify this. Since $D_x^\alpha \rho_\varepsilon(x - y) = (-1)^{|\alpha|} D_y^\alpha \rho_\varepsilon(x - y)$, the weak identity in y transfers derivatives to u and cancels the signs.

For $f = D^\alpha u$, multiply by a cutoff equal to one near $\overline{V}$ and extend as an L^p function to $\mathbb{R}^n$. Minkowski gives

$$\|\rho_\varepsilon * f - f\|_{L^p(V)} \le \int \rho(z) \|f(\cdot - \varepsilon z) - f\|_{L^p(V)} \, dz.$$

Translations are continuous in L^p for finite p: prove this first for continuous compactly supported functions by uniform continuity, then use their L^p density and the translation isometry. The integrand is dominated by $2\|f\|_p$, so dominated convergence applies. Sum the finitely many derivative errors. □

5.3.2 Smooth in the interior (Evans §5.3.2, Theorem 2)

Theorem 5.8 (Meyers–Serrin approximation)
For any open U and finite p, $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$.

Expanded proof.

Choose a locally finite smooth partition (ζ_j) with $\sum_j \zeta_j = 1$ on U, each supported compactly in a relatively compact patch A_j. Slight enlargements of these patches can also be chosen locally finite, by using layers of an exhaustion of U. Extend $f_j = \zeta_j u$ by zero. There are no interface derivatives: the cutoff vanishes in a neighborhood of the edge of its patch.

Given $\delta > 0$, choose radii ε_j so small that

$$\operatorname{supp}(\rho_{\varepsilon_j} * f_j) \subset A_j, \qquad \|\rho_{\varepsilon_j} * f_j - f_j\|_{W^{k,p}(\mathbb{R}^n)} < \delta 2^{-j}.$$

Then $v = \sum_j \rho_{\varepsilon_j} * f_j$ is locally a finite sum of smooth functions. The error series converges absolutely in the Banach space $W^{k,p}$, with norm at most δ. Locally it equals $v - u$, since $\sum_j f_j = u$ locally. Uniqueness of distributions identifies its global sum with $v - u$. Thus $\|v - u\|_{W^{k,p}(U)} \leq \delta$. □

5.3.3 Smooth up to the boundary (Evans §5.3.3, Theorem 3)

Theorem 5.9
If U is bounded with C^1 boundary and $p < \infty$, restrictions of functions smooth in a neighborhood of $\overline{U}$ are dense in $W^{k,p}(U)$.

Expanded proof.

Use a finite partition on $\overline{U}$, subordinate to interior patches and boundary graph patches. In a rotated boundary patch write $U = \{x_n > g(x')\}$ with $g \in C^1$. On the compact base, g is Lipschitz with a constant L. Localize to $f = \zeta u$ supported away from the lateral edges. Translate inward: $f_\delta(x) = f(x + \delta e_n)$.

For $x_n \geq g(x')$ and $|z| < c\delta$,

$$x_n + \delta - z_n - g(x' - z') \geq \delta - (1 + L)|z| > 0 \quad \text{if } c < 1/(1 + L).$$

Thus a small convolution of f_δ is defined in a neighborhood of the closed patch. The translated weak derivatives are $D^\alpha f(x + \delta e_n)$, by a change of test variables. Extend each $D^\alpha f$ by zero only for estimating L^p translations. Translation continuity proves $f_\delta \to f$ in $W^{k,p}$ on the original patch intersected with U. For each fixed δ, choose a smaller convolution radius

每块半径不同

靠近边界的块需要更小的 ε_j。误差预算 $\delta 2^{-j}$ 控制无穷多块的总误差；局部有限性保证内点附近只相加有限项。

giving any desired Sobolev error. The result is smooth across the original boundary. Approximate interior patches by ordinary convolution and sum the finitely many pieces. □

Why the endpoint $p = \infty$ is excluded. The derivative of $|x|$ jumps from -1 to 1. No continuous derivative can approximate it in essential supremum norm with error less than one: the two one-sided requirements contradict continuity at zero. Smooth functions need not be norm dense in $W^{1,\infty}$. Local smoothing does converge in every finite $W^{1,q}$ norm on bounded interior sets and remains uniformly bounded in $W^{1,\infty}$. Later endpoint arguments use those facts instead.

两种稠密性不同

$C^\infty(\overline{U})$ 稠密不等于 $C_c^\infty(U)$ 稠密。常数 1 可光滑逼近，但在有界光滑域中不属于 $W_0^{1,p}$；它的迹不是零。

5.4 Extensions

5.4.1 Reflection and flattening (Evans §5.4, Theorem 1)

Theorem 5.10 (Extension)
Let U be bounded with C^1 boundary, and fix V with $U \Subset V$. For $1 \le p \le \infty$ there is a linear operator

$$E : W^{1,p}(U) \longrightarrow W^{1,p}(\mathbb{R}^n), \qquad Eu = u \text{ on } U,$$

with support in a fixed compact subset of V and $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \le C\|u\|_{W^{1,p}(U)}$. The constant depends on U, V, n, p.

Expanded proof.

先比较两种延拓

反射延拓在接口两侧保持同一函数值；补零只有在迹为零时才不会制造跳跃。延拓算子要同时控制函数和梯度，不能只保持 L^p 范数。

1. A flat boundary. On the upper half-space, for a smooth localized function define

$$\tilde{u}(x', t) = u(x', |t|).$$

For $t < 0$, its tangential derivatives are $(\partial_i u)(x', -t)$ and its normal derivative is $-(\partial_n u)(x', -t)$. Integrating against a test function separately in $t > 0$ and $t < 0$, the interface terms cancel because the values at $t = 0$ agree. Thus these piecewise derivatives are the weak derivatives. For finite p, reflection doubles both the p-th power of the L^p norm and the gradient integral; for $p = \infty$ it preserves their bounds.

2. Flatten a graph. In a boundary chart let $U = \{x_n > g(x')\}$ and set

$$\Phi(y', y_n) = (y', y_n + g(y')).$$

The map and its inverse have bounded first derivatives on the smaller chart, and $\det D\Phi = 1$. For smooth u, $D(u \circ \Phi) = (D\Phi)^T (Du) \circ \Phi$, so $\|u \circ \Phi\|_{W^{1,p}} \le C\|u\|_{W^{1,p}}$, with an analogous inverse bound. For Sobolev u, interior smooth approximation proves the same formula locally in the chart; the global integrability bounds then identify its weak derivatives on the entire chart. Reflect $u \circ \Phi$ and compose with Φ^{-1}.

3. Localize and assemble. Take finitely many smaller boundary charts and an interior patch covering $\overline{U}$, with a smooth partition (ζ_j). Extend each $\zeta_j u$ by the preceding construction, multiplying by a fixed exterior cutoff supported in its chart. The interior term extends by zero. Sum the pieces. On U the sum equals u, because $\sum_j \zeta_j = 1$. The product rule and finitely many chart estimates give the asserted norm bound. Choose all exterior cutoffs inside V. This construction is linear and uses fixed charts and cutoffs.

For finite p, one can also construct first on smooth functions and complete using Theorem 5.9. For $p = \infty$, apply the same reflection formula directly. Locally approximate in finite $W^{1,q}$ norms and pass to the weak identity; the candidate derivatives have the uniform L^∞ bounds already calculated. No $W^{1,\infty}$ norm density is needed. □

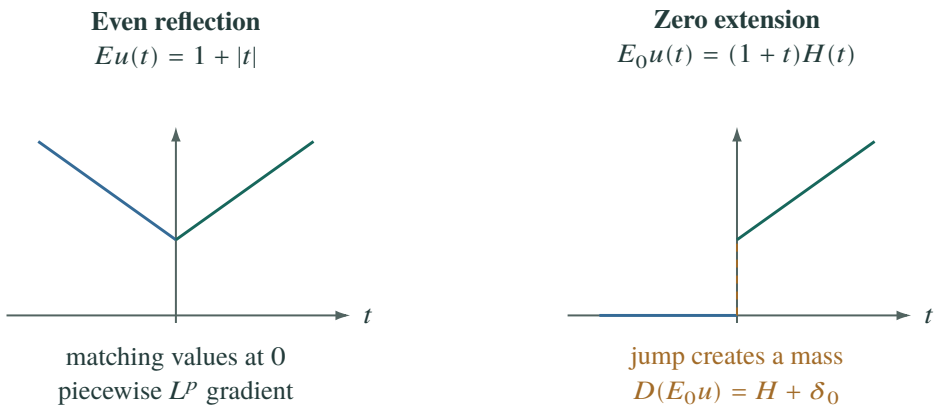


Diagram T5.2. Reflection and zero extension treat the interface differently. Take $u(t) = 1 + t$ for $0 < t < 1$ and inspect only the interface at zero. Reflection matches values and preserves first-order Sobolev regularity locally. Zero extension inserts a jump and a Dirac mass in the first derivative. Even reflection still has a corner, so higher-order extensions need additional matching.

边界正则性用在哪里

C^1 使边界可拉平，坐标变换及逆变换的一阶导数有界；这才保证 Sobolev 范数不被放大到无穷。尖裂缝域不能直接套用此结论。

5.4.2 Higher derivatives: match the jets

Ordinary even reflection need not preserve $W^{2,p}$: for $u(t) = t$ on $t > 0$ it gives $|t|$, whose second derivative has a Dirac mass. For an integer $k \ge 1$,

on $t < 0$ use instead

$$\tilde{u}(x', t) = \sum_{j=1}^{k} a_j u(x', -jt), \qquad \sum_{j=1}^{k} a_j (-j)^{\ell} = 1 \quad (0 \le \ell < k).$$

The Vandermonde matrix is invertible because $-1, \dots, -k$ are distinct. The derivatives of normal order $0, \dots, k-1$ match at zero, so repeated integration by parts has no interface distributions through order k. Each derivative on the lower side is bounded by a finite sum of scaled upper-side derivatives. This proves a $W^{k,p}$ bound on a sufficiently small strip. For $k = 2$ the formula is $3u(x', -t) - 2u(x', -2t)$.

If ∂U is C^k, flattening has bounded derivatives through order k on the chosen compact charts. The higher chain rule and the same partition construction produce an operator bounded simultaneously on $W^{j,p}$ for $0 \le j \le k$. For finite p, smooth boundary approximation passes the identities to the completion; for $p = \infty$, finite-exponent local convergence and the uniform derivative bounds give the same conclusion. We will use a single such operator on L^2 and H^2 in the time-dependent estimates.

高阶延拓不能照抄偶反射

一阶只需接口函数值匹配；二阶还需一阶导数匹配。系数 $3, -2$ 同时满足 $3 - 2 = 1$ 和 $-3 + 4 = 1$，正是为了消掉接口奇异项。

5.5 Traces

5.5.1 Boundary values as a continuous operator

Theorem 5.11 (Evans §5.5, Theorem 1)
If U is bounded with C^1 boundary and $1 \le p < \infty$, there is a unique bounded linear map

$$T : W^{1,p}(U) \longrightarrow L^p(\partial U)$$

agreeing with restriction for functions in $W^{1,p}(U) \cap C(\overline{U})$. In particular, $\|Tu\|_{L^p(\partial U)} \le C \|u\|_{W^{1,p}(U)}$.

Expanded proof.

边界不能随意赋值

改变 u 在零测集上的值不改变 Sobolev 元素；边界通常正是体积零测集。迹由域内逼近的极限决定，不能通过任意指定边界代表值来定义。

1. A flat strip estimate. For smooth v on a strip $Q' \times [0,a]$, the fundamental theorem gives

$$|v(x',0)|^p \le |v(x',t)|^p + p\int_0^a |v|^{p-1}|\partial_n v|(x',s)\,ds.$$

For $p = 1$, use the absolutely continuous function $|v|$ and its derivative bound $|(|v|)'| \le |v'|$. Average t over $(0,a)$ and integrate x'. For $p > 1$, Young's inequality bounds $p|v|^{p-1}|\partial_n v|$ by $C_p(|v|^p + |Dv|^p)$. Consequently

$$\int_{Q'} |v(x',0)|^p\,dx' \le C_{p,a}\int_{Q'\times(0,a)} (|v|^p + |Dv|^p)\,dx.$$

Boundary flattening introduces only bounded Jacobian and derivative factors, including $\sqrt{1+|Dg|^2}$ for surface measure. A finite partition yields the global estimate for smooth u.

2. Complete the restriction map. Choose u_j smooth near $\overline{U}$, with $u_j \to u$ in $W^{1,p}$. The estimate on $u_j - u_l$ shows that $u_j|_{\partial U}$ is Cauchy in $L^p(\partial U)$. Define Tu to be its limit. The same estimate on two approximating sequences proves independence of choice and boundedness. Linearity and uniqueness follow from density.

3. Agreement with continuous functions. For u continuous on $\overline{U}$, the inward-translation and smoothing construction in §5.3 can be chosen to converge uniformly as well as in $W^{1,p}$. Uniform continuity on the compact closure justifies this assertion for each localized piece. The boundary restrictions therefore converge to $u|_{\partial U}$ uniformly, identifying the trace. □

5.5.2 Zero trace and the closure definition

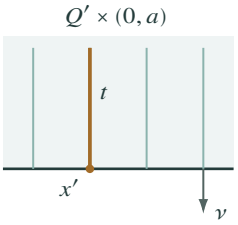
> **Theorem 5.12** (Evans §5.5, Theorem 2)
> *Under the same hypotheses,*
>
> $$W_0^{1,p}(U) = \{u \in W^{1,p}(U) : Tu = 0\}.$$

Expanded proof.

If $u_j \in C_c^\infty(U)$ tends to u in $W^{1,p}$, boundedness of T gives $Tu = \lim Tu_j = 0$.

For the reverse direction we supply the boundary-layer argument. In a flat chart the trace equals the value at $t = 0$ of the absolutely continuous

图 T5.M：沿法向控制边界值



先沿固定 x' 的线段比较内部值与边界值，再对高度和切向变量积分。图中外法向朝下。

迹估计的实际用处

只要 $u_j \to u$ 于 $W^{1,p}$，就自动有 $Tu_j \to Tu$ 于边界 L^p。边界条件因此可随强逼近传递，而不要求逐点追踪边界值。

representative on almost every vertical line. To justify this, use smooth approximations, Fubini, and the one-dimensional estimate

$$|w(0)| \le a^{-1}\|w\|_{L^1(0,a)} + \|w'\|_{L^1(0,a)}.$$

A subsequence converges in $W^{1,p}$ on almost every line, and this estimate identifies the endpoint limit with the trace limit.

Thus if $Tv = 0$, $v(x',t) = \int_0^t \partial_n v(x',s)\,ds$ on almost every such line. Hölder (or the integral triangle inequality for $p = 1$) gives

$$\int_{Q'} \int_0^{2\delta} |v(x',t)|^p \,dt\,dx' \le (2\delta)^p \int_{Q'} \int_0^{2\delta} |\partial_n v(x',t)|^p \,dt\,dx'.$$

Choose a smooth $\chi_\delta(t)$ equal to zero for $t < \delta$, one for $t > 2\delta$, with $|\chi_\delta'| \le C/\delta$. Then

$$D(\chi_\delta v) - Dv = (\chi_\delta - 1)Dv + vD\chi_\delta.$$

The first term tends to zero by absolute continuity of the integral. The p-th power of the second norm is bounded by $C\int_{0<t<2\delta} |Dv|^p$, hence also tends to zero. The function error tends to zero in the same way. Flattening and a finite partition transfer this argument to U. It gives functions supported a positive distance from ∂U converging to u in $W^{1,p}$. For each one, convolution with a smaller radius produces an element of $C_c^\infty(U)$ with arbitrarily small additional error. A diagonal choice completes the proof. □

截断产生的危险项

$D(\chi_\delta u)$ 中有 $uD\chi_\delta$，而 $D\chi_\delta$ 约为 $1/\delta$。零迹给出的法向积分公式补回一个 δ 因子，才使此项趋于零。

Zero extension. If $u \in W_0^{1,p}(U)$, extend compactly supported approximations by zero. They are Cauchy in $W^{1,p}(\mathbb{R}^n)$ with exactly the same norms, so completeness gives a zero extension $\tilde{u}$ satisfying $D\tilde{u} = \widetilde{Du}$. This argument works without regularity of U for finite p. A nonzero trace generally produces a boundary distribution instead.

5.6 Sobolev Inequalities

5.6.1 Scaling predicts the exponent

For $u_\lambda(x) = u(\lambda x)$ on $\mathbb{R}^n$,

$$\|u_\lambda\|_q = \lambda^{-n/q}\|u\|_q, \qquad \|Du_\lambda\|_p = \lambda^{1-n/p}\|Du\|_p.$$

An estimate $\|u\|_q \le C\|Du\|_p$ with a constant independent of scale must therefore have $1/q = 1/p - 1/n$. For $p < n$ put $p^* = np/(n-p)$.

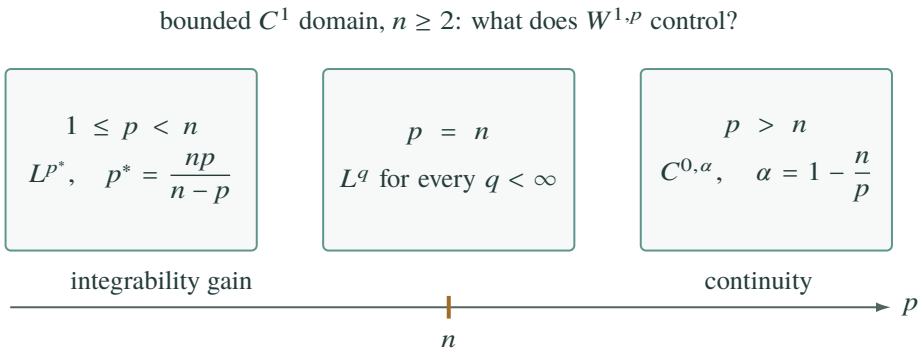


Diagram T5.3. The dimension separates three embedding regimes. The diagram records standard continuous embeddings on a bounded C^1 domain with $n \geq 2$. At $p = n$, the conclusion is valid for each finite q, with constants that may depend on q. It does not provide a uniform estimate as $q \to \infty$. Compactness is a further question addressed in the next section.

5.6.2 Gagliardo–Nirenberg–Sobolev (Evans §5.6.1)

We first give the integral inequality behind the endpoint $p = 1$. For $n \geq 2$ and nonnegative f_i on $\mathbb{R}^{n-1}$, with $\hat{x}_i$ denoting all coordinates except x_i,

$$\int_{\mathbb{R}^n} \prod_{i=1}^{n} f_i(\hat{x}_i)^{1/(n-1)}\,dx \leq \prod_{i=1}^{n} \left(\int_{\mathbb{R}^{n-1}} f_i \right)^{1/(n-1)}. \tag{LW}$$

Expanded proof (editorial integral lemma).

For $n = 2$ this is Fubini. Suppose it holds in dimension $n-1$. Integrate first in x_n and apply Hölder with $n-1$ equal exponents to the factors $i < n$. Put $F_i = \int_{\mathbb{R}} f_i(\hat{x}_i)\,dx_n$. What remains is

$$\int_{\mathbb{R}^{n-1}} f_n^{1/(n-1)} \prod_{i<n} F_i^{1/(n-1)}\,dx'.$$

Apply Hölder with exponents $n-1$ and $(n-1)/(n-2)$, obtaining

$$\left(\int f_n \right)^{1/(n-1)} \left(\int_{\mathbb{R}^{n-1}} \prod_{i<n} F_i^{1/(n-2)} \right)^{(n-2)/(n-1)}.$$

The induction hypothesis on the last integral gives (LW). Tonelli justifies the order changes, first for finite integrals and then by monotone truncation in general. □

Theorem 5.13 (Evans §5.6.1, Theorem 1)
For $1 \leq p < n$ and $u \in C_c^1(\mathbb{R}^n)$,

$$\|u\|_{p^*} \leq C_{n,p} \|Du\|_p.$$

Expanded proof.

先用缩放验算

$1/p^* = 1/p - 1/n$ 是量纲平衡，不是记忆技巧。若估计两侧缩放次数不同，就必须出现域尺度、低阶项或其他约束。

1. Start with $p = 1$. The fundamental theorem on the ith coordinate line gives

$$|u(x)| \le \int_{\mathbb{R}} |\partial_i u(\hat{x}_i, t)|\, dt =: f_i(\hat{x}_i).$$

Multiply the n inequalities and raise to the power $1/(n-1)$. By (LW),

$$\int |u|^{n/(n-1)} \le \prod_i \|\partial_i u\|_1^{1/(n-1)} \le \|Du\|_1^{n/(n-1)}.$$

Taking the power $(n-1)/n$ proves the endpoint.

2. Obtain $1 < p < n$ by a power substitution. Set $a = p(n-1)/(n-p) > 1$. Apply the endpoint to $|u|^a$, a compactly supported C^1 function. Hölder with $p' = p/(p-1)$ yields

$$\Big(\int |u|^{an/(n-1)}\Big)^{(n-1)/n} \le Ca \int |u|^{a-1}|Du| \le Ca\|Du\|_p \Big(\int |u|^{(a-1)p'}\Big)^{1/p'}.$$

Both exponents $an/(n-1)$ and $(a-1)p'$ equal p^*. The left side is $\|u\|_{p^*}^a$ and the last factor is $\|u\|_{p^*}^{a-1}$. Cancel it if nonzero; if zero the conclusion is immediate. □

Corollary 5.14 (Evans §5.6.1, Theorems 2–3)
On a bounded C^1 domain, $\|u\|_{p^*} \le C\|u\|_{W^{1,p}(U)}$. On any bounded open set, for $u \in W_0^{1,p}(U)$,

$$\|u\|_{p^*} \le C_{n,p}\|Du\|_p, \qquad \|u\|_p \le C_{n,p}|U|^{1/n}\|Du\|_p.$$

Expanded proof.

For the first statement extend by Theorem 5.10, smooth on $\mathbb{R}^n$ with fixed compact support, and apply Theorem 5.13. Applying it to differences shows the smooth sequence is Cauchy in L^{p^*}. Its L^p limit identifies the L^{p^*} limit with the extension, so the inequality passes to u. For zero boundary data use zero extensions of the defining compactly supported sequence instead. Finally Hölder gives $\|u\|_p \le |U|^{1/p-1/p^*}\|u\|_{p^*}$ and $1/p - 1/p^* = 1/n$. □

幂次为何这样选

a 同时满足 $an/(n-1) = p^*$ 与 $(a-1)p' = p^*$，所以最后能约去同一个函数范数。随便取幂次，两个积分指数就对不上。

为何一般域估计有低阶项

常数 $u = 1$ 满足 $Du = 0$，所以对任意 $W^{1,p}$ 函数不能只有梯度控制。零迹排除非零常数，才可去掉右侧的 $\|u\|_p$。

5.6.3 The critical case $p = n$

For $n \ge 2$, bounded C^1 domains satisfy $W^{1,n}(U) \hookrightarrow L^q(U)$ for every finite q. Indeed choose $1 \le r < n$ so close to n that $r^* \ge q$. Finite measure gives $W^{1,n} \subset W^{1,r}$, and the preceding inequality followed by Hölder proves the claim. The constant depends on q and need not stay bounded as $q \to \infty$.

A useful counterexample, equivalent in behavior to source Problem 14, is

$$u(x) = \log\log(e/|x|) \quad (0 < |x| < 1).$$

It is unbounded, but $|Du| = 1/(|x|\log(e/|x|))$ and

$$\int_{B_1} |Du|^n = C_n \int_0^1 \frac{dr}{r\log(e/r)^n} < \infty \quad (n > 1).$$

The function itself has finite L^n norm. Its weak derivative is identified by integration outside B_ε; the interface term is $O(\varepsilon^{n-1}\log\log(e/\varepsilon)) \to 0$. In dimension one, by contrast, $W^{1,1}$ on an interval has an absolutely continuous, hence bounded, representative.

5.6.4 Morrey's inequality (Evans §5.6.2, Theorems 4–5)

Theorem 5.15

If $n < p < \infty$, then $W^{1,p}(\mathbb{R}^n)$ has a representative satisfying

$$|u(x) - u(y)| \le C_{n,p}|x-y|^{1-n/p}\|Du\|_{L^p(\mathbb{R}^n)}, \qquad \|u\|_\infty \le C_{n,p}\|u\|_{W^{1,p}}.$$

On a bounded C^1 domain, $\|u\|_{C^{0,1-n/p}(\overline{U})} \le C\|u\|_{W^{1,p}(U)}$. For $p = \infty$, the corresponding exponent is 1.

Expanded proof.

1. A local kernel estimate. Let D be a ball of radius R and $x \in \overline{D}$. For smooth u, integrate the line identity $|u(x) - u(z)| \le \int_0^{|z-x|} |Du(x + t\omega)|\,dt$ over $z = x + s\omega \in D$. Convexity keeps each segment inside D. The s integration after changing order contributes at most CR^n. Dividing by $|D|$ gives

$$|u(x) - u_D| \le C_n \int_D \frac{|Du(z)|}{|x-z|^{n-1}}\,dz, \qquad u_D = |D|^{-1}\int_D u.$$

Hölder bounds the integral by

$$C\|Du\|_{L^p(D)} \left(\int_0^{2R} t^{n-1-(n-1)p'}\,dt\right)^{1/p'} \le CR^{1-n/p}\|Du\|_{L^p(D)}.$$

The radial integral is finite exactly when $p > n$ (for $n = 1$, the same computation is simply the integral triangle inequality and Hölder).

2. Compare two points. For $r = |x-y|$, choose $D = B(x, 2r)$ containing both. Bound $|u(x) - u_D|$ and $|u(y) - u_D|$ by step 1. This also yields the local estimate

$$|u(x) - u(y)| \le Cr^{1-n/p}\|Du\|_{L^p(B(x,2r))}$$

临界点不是无穷大嵌入

$p \uparrow n$ 时公式中的 p^* 趋于无穷，但常数也可能发散，不能把极限直接代入。$\log\log(e/|x|)$ 给出明确反例。

whenever that ball lies in the domain. With a unit ball in step 1, $|u(x)| \le |u_D| + C\|Du\|_p \le C(\|u\|_p + \|Du\|_p)$.

3. Pass to Sobolev functions. Smooth approximation on $\mathbb{R}^n$ and the inequalities on differences give uniform convergence to a Hölder representative, agreeing almost everywhere with the L^p limit. On U, use the extension operator. For $p = \infty$ on $\mathbb{R}^n$, local smoothing has gradient bounded by $\|Du\|_\infty$; the line identity gives a common Lipschitz constant. A locally uniform subsequential limit, identified by finite L^q convergence, proves the assertion. Extend first for the domain version. □

5.6.5 Higher-order embeddings (Evans §5.6.3, Theorem 6)

Let U be bounded with C^1 boundary, $k \ge 1$, and $1 \le p < \infty$. If $kp < n$,

$$W^{k,p}(U) \hookrightarrow L^q(U), \qquad \frac{1}{q} = \frac{1}{p} - \frac{k}{n}.$$

If $k > n/p$ and n/p is not an integer, put $l = \lfloor n/p \rfloor$, $m = k - l - 1$, and $\gamma = l + 1 - n/p$. Then $W^{k,p} \hookrightarrow C^{m,\gamma}(\overline{U})$. If $n/p = j$ is an integer and $k > j$, the conclusion is $C^{k-j-1,\gamma}$ for every $0 < \gamma < 1$. Each embedding has a norm bound depending on U, n, k, p and, where applicable, γ.

Expanded proof.

Apply the first-order estimate to each derivative of order at most $k-1$. This improves $W^{k,p}$ to W^{k-1,p_1} with $1/p_1 = 1/p - 1/n$. Iterate while the input exponent is below n. If $kp < n$, exactly k steps yield L^q. If n/p is nonintegral, after l steps the exponent r satisfies $1/r = 1/p - l/n$ and $r > n$. Apply Morrey to each derivative through order $k - l - 1$; $1 - n/r = l + 1 - n/p$. If $n/p = j$ is integral, after $j-1$ steps the exponent is n. The critical finite-q estimate gives $W^{k-j,q}$ for arbitrary finite q. Choose $q > n$ and apply Morrey, obtaining $C^{k-j-1,1-n/q}$. Any $\gamma < 1$ follows by choosing q sufficiently large. Multiplying the finitely many estimate constants gives the bounds. The continuous weak derivatives obtained here are classical derivatives: mollify locally and pass to the coordinate-segment integral identity, as in the completeness proof in §5.1. For $n = p = 1$, interval absolute continuity gives the needed bounded derivatives directly and hence the stated weaker Hölder conclusions. □

连续代表元的含义

原先 u 是几乎处处等价类。Morrey 证明其中存在连续函数；两个连续函数若几乎处处相同，便处处相同。因此此时谈逐点值才有唯一含义。

5.7 Compactness

5.7.1 Rellich–Kondrachov (Evans §5.7, Theorem 1)

Theorem 5.16
Let U be bounded with C^1 boundary. If $1 \le p < n$ and $1 \le q < p^*$, every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$. We write $W^{1,p}(U) \Subset L^q(U)$. For $p = n$ the same conclusion holds for every finite q. For $p > n$, including $p = \infty$, bounded sequences have subsequences converging uniformly on $\overline{U}$.

Expanded proof.

1. Put every function on the same compact set. Extend a bounded sequence (u_j) by the fixed operator in Theorem 5.10. Call the extensions v_j. They have uniformly bounded $W^{1,p}(\mathbb{R}^n)$ norm and support in a fixed compact set K.

2. Uniform smoothing error. For smooth v, the line integral formula and Minkowski give

$$\|v(\cdot + h) - v\|_p \le |h| \int_0^1 \|Dv(\cdot + th)\|_p \, dt = |h| \|Dv\|_p.$$

Smooth approximation passes this to every finite-p Sobolev function. Consequently

$$\|\rho_\varepsilon * v_j - v_j\|_p \le C\varepsilon \|Dv_j\|_p \le C\varepsilon$$

uniformly in j. For $q \le p$, finite support gives the corresponding L^q bound. For $p < q < p^*$ choose $0 < \theta < 1$ with $1/q = \theta/p + (1-\theta)/p^*$. Hölder interpolation gives

$$\|\rho_\varepsilon * v_j - v_j\|_q \le \|\rho_\varepsilon * v_j - v_j\|_p^\theta \|\rho_\varepsilon * v_j - v_j\|_{p^*}^{1-\theta} \le C\varepsilon^\theta.$$

The second factor is uniformly bounded by the Sobolev inequality and convolution's L^{p^*} bound.

3. Compactness for a fixed smoothing radius. For fixed ε, the functions $v_{j,\varepsilon} = \rho_\varepsilon * v_j$ have support in $K + \overline{B}_\varepsilon$. Moreover

$$\|v_{j,\varepsilon}\|_\infty \le \|\rho_\varepsilon\|_\infty \|v_j\|_1, \qquad \|Dv_{j,\varepsilon}\|_\infty \le \|D\rho_\varepsilon\|_\infty \|v_j\|_1.$$

The right sides are independent of j by bounded support and Hölder. Arzelà–Ascoli, invoked here as a prerequisite, supplies a uniformly convergent subsequence on that compact set.

高阶指数逐层算

每使用一次一阶嵌入，导数阶数减一，$1/p$ 减去 $1/n$。到达临界指数时先转到任意有限 q，所以整数临界处不能擅自宣称 Lipschitz。

连续嵌入与紧嵌入

连续嵌入只把有界集送到有界集；紧嵌入还能从任意有界序列抽出强收敛子列。强收敛常用于通过非线性项的极限。

4. Make one subsequence work at every scale. Choose $\varepsilon_l \downarrow 0$ so the L^q smoothing error is at most 2^{-l}. Extract nested subsequences for $l = 1, 2, \ldots$, and take the diagonal sequence (v_{j_a}). For fixed l, its smoothed sequence is Cauchy, so

$$\limsup_{a,b\to\infty} \|v_{j_a} - v_{j_b}\|_q \le 2^{1-l}.$$

Let $l \to \infty$. Completeness of L^q gives convergence, and restriction to U proves the result.

5. The remaining exponents. If $p = n \ge 2$, choose $1 \le r < n$ with $r^* > q$ and apply the preceding result to the bounded $W^{1,r}$ sequence. If $p > n$, Morrey gives uniform boundedness and a common Hölder modulus on $\overline{U}$; Arzelà–Ascoli gives uniform convergence. For the exceptional case $n = p = 1$, extension to a fixed bounded interval and the translation/smoothing argument above prove compactness in L^1. The interval fundamental theorem gives a uniform L^∞ bound; interpolation then gives compactness in every finite L^q. □

5.7.2 Why the critical exponent is excluded

Fix nonzero $\psi \in C_c^\infty(B_1)$ and an interior point x_0. For $p < n$ define, for sufficiently large λ,

$$u_\lambda(x) = \lambda^{(n-p)/p}\psi(\lambda(x - x_0)).$$

Then $\|Du_\lambda\|_p = \|D\psi\|_p$, $\|u_\lambda\|_p = \lambda^{-1}\|\psi\|_p$, but $\|u_\lambda\|_{p^*} = \|\psi\|_{p^*} > 0$. The sequence tends to zero at every $x \ne x_0$. A strongly L^{p^*} convergent subsequence would have a further almost-everywhere convergent subsequence, so its limit would be zero, contradicting its constant nonzero norm.

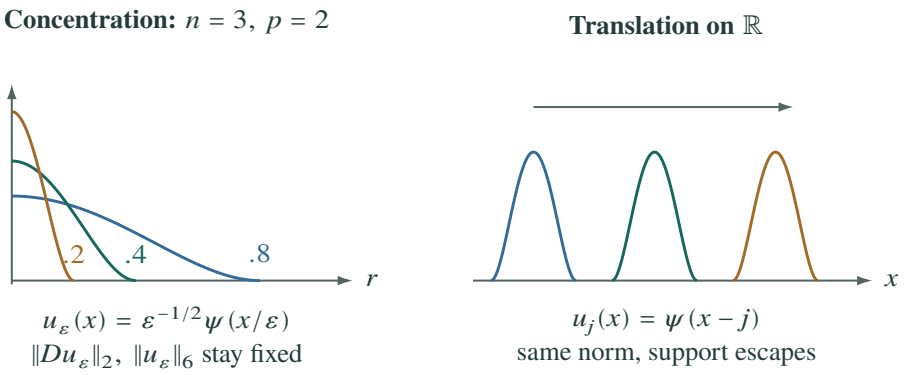


Diagram T5.4. Bounded norms do not prevent concentration or escape. The left panel uses radial bump profiles $\psi(x) = (1 - |x|^2)_+^2$ in $\mathbb{R}^3$, scaled with $\varepsilon = 0.8, 0.4, 0.2$; their gradient L^2 and critical L^6 norms stay fixed. The right panel illustrates translations of a compact bump on $\mathbb{R}$. Each mechanism obstructs strong compactness for a different reason.

对角线到底做什么

固定 ε 得到的子列可能随 ε 改变。先抽嵌套子列再取对角线，保证每个固定尺度最终都收敛；最后才令尺度趋零。

集中导致临界失紧

尖峰越来越窄、越来越高，梯度范数和临界范数都不变。次临界范数会衰减，因此 $q < p^*$ 留出的余量正好压住集中。

Zero boundary data on rough bounded domains. For finite p, zero extensions of a bounded sequence in $W_0^{1,p}(U)$ have a common compact support and bounded whole-space Sobolev norm. The proof above therefore gives compactness in $L^p(U)$ on any bounded open U, without regular boundary. On unbounded domains, translations of a fixed bump can escape to infinity and destroy global compactness.

5.8 Additional Topics

5.8.1 Poincaré inequalities (Evans §5.8.1, Theorems 1–2)

Lemma 5.17 (Zero gradient means constant)
If U is connected and $u \in W^{1,1}_{\mathrm{loc}}(U)$ satisfies $Du = 0$, then u is constant almost everywhere.

Expanded proof (also source Problem 11).

On a ball compactly contained in U, mollification has zero classical gradient on each smaller concentric ball and is constant there. As $\varepsilon \to 0$ it converges in L^1; the constants converge to the average of u on the smaller ball. Hence u is constant there almost everywhere. Constants agree on overlapping balls, because the overlap has positive measure. Any two points in an open connected subset of $\mathbb{R}^n$ can be joined by a finite chain of such balls: the set of points reachable from a fixed ball is both relatively open and relatively closed. Connectedness completes the proof. □

Theorem 5.18
On a bounded connected C^1 domain, for $1 \le p \le \infty$,

$$\|u - u_U\|_{L^p(U)} \le C_{U,n,p}\|Du\|_{L^p(U)}, \qquad u_U = |U|^{-1}\int_U u.$$

On a ball B_r the constant is $C_{n,p}r$.

连通性不可删

在两个不相交的球上分别取常数 0 和 1，梯度仍处处为零，却不是同一个常数。均值型 Poincaré 必须排除这种情况。

Expanded proof.

If the inequality failed, normalization would give

$$(v_j)_U = 0, \quad \|v_j\|_p = 1, \quad \|Dv_j\|_p < 1/j.$$

For finite p, compactness gives a subsequence converging strongly in L^p to v; for $p = \infty$, use the uniform compactness just proved. The mean and norm pass to the limit, giving $v_U = 0$ and $\|v\|_p = 1$. Testing the weak derivative identity and passing to the limit gives $Dv = 0$. Lemma 5.17 makes v constant, and its zero mean makes it zero, a contradiction. For

$B_r(x_0)$ use $w(y) = u(x_0 + ry)$: the function norm scales by $r^{-n/p}$ and the gradient norm by $r^{1-n/p}$, leaving precisely one factor r. □

For $u \in W^{1,n}(\mathbb{R}^n)$, $n \geq 2$, the $p = 1$ ball estimate and Hölder give

$$\frac{1}{|B_r|}\int_{B_r} |u - u_{B_r}| \leq Cr\frac{1}{|B_r|}\int_{B_r} |Du| \leq C\|Du\|_{L^n(\mathbb{R}^n)}.$$

Thus $W^{1,n}$ controls bounded mean oscillation (BMO), whose seminorm is the supremum of the left side over balls. BMO controls average oscillation rather than pointwise size.

5.8.2 Difference quotients (Evans §5.8.2, Theorem 3)

For $h \neq 0$ define

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}.$$

Theorem 5.19

If $u \in W^{1,p}(U)$, $V \Subset U$, and $|h| < \frac{1}{2}\operatorname{dist}(V, \partial U)$, then $\|D_i^h u\|_{L^p(V)} \leq \|\partial_i u\|_{L^p(U)}$ for finite p. Conversely, if $1 < p < \infty$, $u \in L^p(U)$, and all these quotients are uniformly bounded in $L^p(V)$ for every coordinate, then $u \in W^{1,p}(V)$, with the corresponding derivative bounds.

Expanded proof.

For smooth u, $D_i^h u(x) = \int_0^1 \partial_i u(x + the_i)\, dt$. Jensen and Fubini give

$$\int_V |D_i^h u|^p \leq \int_0^1 \int_V |\partial_i u(x + the_i)|^p \, dx\, dt \leq \int_U |\partial_i u|^p.$$

Interior approximation on a slightly larger set passes this to u.

For the converse, choose $h_j \to 0$. Reflexivity of $L^p(V)$ gives, after a finite diagonal extraction over coordinates, $D_i^{h_j} u \rightharpoonup g_i$ in $L^p(V)$. For a test function ϕ and small h a change of variables yields

$$\int D_i^h u\, \phi = -\int u D_i^{-h}\phi.$$

There are no edge terms because the test support and its translates stay inside V. Since $D_i^{-h}\phi \to \partial_i \phi$ uniformly on a common compact support, the right side tends to $-\int u\partial_i\phi$. The left side tends weakly to $\int g_i\phi$. Thus $g_i = \partial_i u$. Weak lower semicontinuity proves the norm bound. □

消掉常数的两种方式

零迹型要求 $Tu = 0$；均值型减去 u_U。二者都排除梯度算子的常数核，但适用条件和边界信息不同。

For finite p, one also has $D_i^h u \to \partial_i u$ strongly on interior sets: subtract the derivative in the line-average formula and use L^p translation continuity. This gives a practical way to recognize weak derivatives in elliptic regularity estimates.

5.8.3 Lipschitz functions (Evans §5.8.2, Theorem 4)

Locally, $W^{1,\infty}$ functions have Lipschitz representatives, by smoothing on balls and using the gradient bound on segments. Conversely an L-Lipschitz function has bounded difference quotients. On a relatively compact ball, extract weak limits in L^2. The weak identity above gives weak derivatives; the limit still satisfies $|g_i| \le L$ almost everywhere, since the closed convex set $\{g : |g| \le L\}$ is weakly closed in L^2. This identifies locally bounded weak derivatives without appealing to Rademacher's theorem.

On bounded C^1 domains the extension theorem gives a global Lipschitz representative for $W^{1,\infty}(U)$, with constant controlled by the full $W^{1,\infty}$ norm. Global Lipschitz continuity need not follow on arbitrary domains: points across a narrow slit can be very close in Euclidean distance but far apart along paths inside the domain.

$p = 1$ 的反向结论失效

阶跃函数的差商高度为 $1/|h|$、支集宽约 $|h|$，故 L^1 范数有界；极限却是 Dirac 质量。L^1 有界不保证弱子列极限仍是函数。

5.8.4 Differentiability a.e. (Evans §5.8.3, Theorems 5–6)

Theorem 5.20

For $p > n$, a function $u \in W^{1,p}_{\mathrm{loc}}(U)$, identified with its continuous representative, is differentiable almost everywhere, and its classical gradient equals its weak gradient there. Every locally Lipschitz function is therefore differentiable a.e.

避免循环证明

先由差商得到 Lipschitz 函数的弱导数，再由 Morrey 和微分定理证明几乎处处可微。若先引用 Rademacher 来构造弱导数，又用它证明 Rademacher，就循环了。

Expanded proof.

First let $n < p < \infty$. The L^p version of the Lebesgue differentiation theorem, invoked here, says that for almost every x,

$$\frac{1}{|B_r|}\int_{B_r(x)} |Du(z) - Du(x)|^p \, dz \to 0.$$

Fix such an x and put $v(z) = u(z) - u(x) - Du(x)\cdot(z-x)$. For $r = |y-x|$ small enough that $B(x,2r) \Subset U$, the local Morrey estimate gives

$$\begin{aligned} |v(y) - v(x)| &\le C r^{1-n/p} \left(\int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} \\ &= C r \left(\frac{1}{|B_{2r}|} \int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} = o(r). \end{aligned}$$

This is precisely differentiability with derivative $Du(x)$. For $p = \infty$, work locally with any finite exponent greater than n. A locally Lipschitz function belongs locally to $W^{1,\infty}$ by the preceding argument, proving Rademacher's theorem as a consequence. □

5.8.5 Hardy's inequality (Evans §5.8.4, Theorem 7)

Theorem 5.21
If $n \geq 3$ and $u \in H^1(B_r)$, then

$$\int_{B_r} \frac{|u|^2}{|x|^2}\,dx \leq C_n \int_{B_r} \left(|Du|^2 + r^{-2}|u|^2\right)\,dx.$$

Expanded proof.

Start with u smooth on $\overline{B}_r$ and set $X = x/|x|^2$, so $\operatorname{div} X = (n-2)/|x|^2$. Apply the divergence theorem on $B_r \setminus \overline{B}_\varepsilon$. The inner boundary term is $O(\varepsilon^{n-2})$ and tends to zero. With $I = \int_{B_r} u^2/|x|^2$ and $A = \|Du\|_2$, the result is

$$(n-2)I = r^{-1}\int_{\partial B_r} u^2\,dS - 2\int_{B_r} u\,Du \cdot \frac{x}{|x|^2}.$$

Cauchy–Schwarz and Young yield $I \leq C_n A^2 + C_n r^{-1}\int_{\partial B_r} u^2$. To control the last term, apply divergence to xu^2:

$$r\int_{\partial B_r} u^2 = \int_{B_r} (nu^2 + 2u\,x \cdot Du) \leq C_n \int_{B_r} (u^2 + r^2|Du|^2).$$

Combine the estimates. For general u, use smooth boundary approximation. Applying the estimate to differences shows $u_j/|x|$ is Cauchy in L^2. An a.e. convergent subsequence identifies its limit with $u/|x|$, and the bound passes to the limit. □

5.8.6 Fourier characterization (Evans §5.8.5, Theorem 8)

Use the unitary convention $\hat{u}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi}u(x)\,dx$. For integer $k \geq 0$,

$$u \in H^k(\mathbb{R}^n) \iff (1+|\xi|^2)^{k/2}\hat{u} \in L^2(\mathbb{R}^n),$$

$$\|u\|_{H^k} \asymp \|(1+|\xi|^2)^{k/2}\hat{u}\|_2.$$

Expanded proof.

For compactly supported smooth functions integration by parts gives $\widehat{D^\alpha u} = (i\xi)^\alpha \hat{u}$. Whole-space cutoff and convolution approximate every

可微比 Hölder 更强

Morrey 单独给出 $O(r^{1-n/p})$。减去候选线性项后，梯度的局部平均误差趋零，才把余项改善为 $o(r)$。

奇点与边界都要处理

不能直接在含原点的球上对奇异向量场用散度定理；先挖去小球再估内边界项。外边界项不为零，所以一般球域估计保留 $r^{-2}\|u\|_2^2$。

H^k function by such functions. Plancherel and passage to the L^2 limit preserve the identity. Also

$$c_{n,k}(1+|\xi|^2)^k \le \sum_{|\alpha|\le k} |\xi^\alpha|^2 \le C_{n,k}(1+|\xi|^2)^k.$$

This follows by the multinomial expansion of $(1+\xi_1^2+\cdots+\xi_n^2)^k$ and comparison of its finitely many positive coefficients. It proves the forward bound. Conversely, the weighted condition makes $g_\alpha = \mathscr{F}^{-1}((i\xi)^\alpha \hat{u})$ an L^2 function. Plancherel against test functions gives the weak derivative identity, hence $g_\alpha = D^\alpha u$ and $u \in H^k$. The same polynomial comparison proves both norm bounds. □

For real $s > 0$ this suggests the definition

$$\|u\|_{H^s(\mathbb{R}^n)} = \|(1+|\xi|^2)^{s/2}\hat{u}\|_2.$$

It is equivalent to the source weight $1+|\xi|^s$; equivalent norms describe the same space.

频率中的光滑度

求导变成乘以 $i\xi$。高频尾部越小，可承受的频率权重越高，意味着可积的导数越多；分数阶空间沿用同一机制。

5.9 Other Spaces of Functions

5.9.1 The dual space H^{-1} (Evans §5.9.1, Theorem 1)

Define $H^{-1}(U) = (H_0^1(U))^*$ with

$$\|f\|_{H^{-1}} = \sup_{\|v\|_{H^1}\le 1} |\langle f, v\rangle|.$$

Here we use the full H^1 norm, rather than the equivalent gradient norm available on bounded domains.

Theorem 5.22

For every $f \in H^{-1}(U)$ there are $f^0, \dots, f^n \in L^2(U)$ with

$$\langle f, v\rangle = \int_U f^0 v + \sum_{i=1}^n f^i \partial_i v \, dx, \qquad v \in H_0^1(U).$$

Moreover

$$\|f\|_{H^{-1}} = \min\left\{ \left(\sum_{i=0}^n \|f^i\|_2^2 \right)^{1/2} : (f^0, \dots, f^n) \text{ represents } f \right\}.$$

In distributions, $f = f^0 - \sum_i \partial_i f^i$.

Expanded proof.

Apply the Hilbert-space Riesz representation theorem, an external functional analysis result, to $H_0^1(U)$ with inner product $\int (wv + Dw \cdot Dv)$. There is a unique $w \in H_0^1(U)$ such that

$$\langle f, v\rangle = \int_U (wv + Dw \cdot Dv), \qquad \|f\|_{H^{-1}} = \|w\|_{H^1}.$$

Set $f^0 = w$ and $f^i = \partial_i w$. Their product L^2 norm equals $\|f\|_{H^{-1}}$. Any other representation (g^i) satisfies $|\langle f, v\rangle| \le (\sum_i \|g^i\|_2^2)^{1/2}\|v\|_{H^1}$ by Cauchy–Schwarz. Take the supremum over unit v to prove minimality. The distributional sign follows from $\langle \partial_i g, v\rangle = -\int g\partial_i v$. □

The embedding $L^2(U) \to H^{-1}(U)$ is $\langle f, v\rangle = \int_U fv$, with norm at most $\|f\|_2$. It is injective because test functions separate locally integrable functions. Thus the useful pivot-space chain is

$$H_0^1(U) \subset L^2(U) \subset H^{-1}(U).$$

负一阶不是负次求导

H^{-1} 的元素可以是导数型分布。若 $F \in L^2(U;\mathbb{R}^n)$，则 $-\operatorname{div} F$ 作用在 v 上是 $\int F \cdot Dv$，由 Cauchy 不等式连续。

5.9.2 Banach-valued time functions (Evans §5.9.2, Theorem 2)

For a real Banach space X, $L^p(0,T;X)$ consists of strongly measurable functions with integrable pth power of their X norm (or essentially bounded norm when $p = \infty$). Integration below is Bochner integration. Its construction, norm inequality, and Fubini properties are prerequisites from measure theory. We say $u' = g$ weakly if

$$\int_0^T u(t)\phi'(t)\,dt = -\int_0^T g(t)\phi(t)\,dt \quad (\phi \in C_c^\infty(0,T)),$$

as an equality in X. The space $W^{1,p}(0,T;X)$ requires both u and g in $L^p(0,T;X)$.

Theorem 5.23 (Fundamental theorem in X)
Every $u \in W^{1,p}(0,T;X)$ has a continuous representative with

$$u(t) - u(s) = \int_s^t u'(\tau)\,d\tau.$$

$$\max_{[0,T]} \|u(t)\|_X \le T^{-1/p}\|u\|_{L^p(X)} + T^{1-1/p}\|u'\|_{L^p(X)}.$$

Interpret $1/p = 0$ for $p = \infty$.

Expanded proof.

Set $G(t) = \int_0^t u'(\tau)\,d\tau$. The Bochner integral norm inequality makes G absolutely continuous. Fubini against scalar test functions shows $G' = u'$ weakly, hence $w = u - G$ has weak derivative zero.

不要抹去中间的 L^2

Riesz 可把 Hilbert 空间与其对偶抽象同构，但演化方程要保留 L^2 配对。u_t 可只在 H^{-1} 中，$\langle u_t, u\rangle$ 仍有意义。

To show w is constant without making an uncountable exceptional-set argument over X^*, smooth w in time on interior intervals. Its smooth derivative is zero, so w_ε is an X-valued constant there. The convergence $w_\varepsilon \to w$ in local $L^1(0,T;X)$ forces these constants to converge in X on each fixed subinterval. Overlapping subintervals give the same constant c. Thus $u = c + G$ almost everywhere; take $c + G$ as the representative. The formula holds for all s, t, including endpoints. Its norm inequality gives

$$\|u(t)\|_X \le \|u(s)\|_X + \int_0^T \|u'(\tau)\|_X \, d\tau.$$

Average in s and apply Hölder, obtaining the stated bound. □

5.9.3 Energy-space continuity (Evans §5.9.2, Theorem 3)

Write $V = H_0^1(U)$, $H = L^2(U)$, and $V^* = H^{-1}(U)$.

Theorem 5.24
If $u \in L^2(0,T;V)$ and $u' \in L^2(0,T;V^*)$, then u has a representative in $C([0,T];H)$ and

$$\|u(t)\|_H^2 - \|u(s)\|_H^2 = 2\int_s^t \langle u'(\tau), u(\tau)\rangle \, d\tau.$$

The squared norm is absolutely continuous and its derivative is the integrand almost everywhere. Also

$$\max_t \|u(t)\|_H \le C_T\big(\|u\|_{L^2(V)} + \|u'\|_{L^2(V^*)}\big).$$

Expanded proof.

1. Construct time-smooth approximations without endpoint jumps. Since $V \subset V^*$ continuously, $u \in W^{1,2}(0,T;V^*)$. Theorem 5.23 gives a continuous V^* representative. Reflect it across 0 and T to a slightly larger interval. The reflected function remains in $L^2(V)$, with weak derivative in $L^2(V^*)$ and the appropriate reflected signs. There is no endpoint Dirac term because the V^* values match. Convolve in time. Translation continuity of Bochner L^2 gives

$$u_\varepsilon \to u \text{ in } L^2(V), \qquad u_\varepsilon' \to u' \text{ in } L^2(V^*).$$

Each smoothed function is smooth with values in V.

2. A uniform-in-time Cauchy estimate. For any smooth $w : [0,T] \to V$, $\frac{d}{dt}\|w\|_H^2 = 2\langle w', w\rangle$. Integrate between s and t, take absolute values of

初值需要时间代表元

L^p 时间函数可以在单个时刻任意修改，因此 $u(0)$ 原先没有确定意义。时间基本定理选出连续代表元，才允许严谨地指定初值。

the integral, and average s over $(0,T)$. It follows that

$$\sup_t \|w(t)\|_H^2 \le T^{-1}\|w\|_{L^2(H)}^2 + 2\|w'\|_{L^2(V^*)}\|w\|_{L^2(V)}.$$

Apply this to $w = u_\varepsilon - u_\delta$. Both graph-norm errors tend to zero, so (u_ε) is Cauchy in $C([0,T];H)$. Its uniform limit agrees with u almost everywhere by the $L^2(H)$ convergence. This proves continuity, including at both endpoints.

3. Pass the energy identity to the limit. The endpoint terms converge uniformly. The integrands converge in L^1 because

$$\begin{aligned}\|\langle u_\varepsilon', u_\varepsilon\rangle - \langle u', u\rangle\|_{L^1} &\le \|u_\varepsilon' - u'\|_{L^2(V^*)}\|u_\varepsilon\|_{L^2(V)} \\ &\quad + \|u'\|_{L^2(V^*)}\|u_\varepsilon - u\|_{L^2(V)} \to 0.\end{aligned}$$

The limiting integral identity gives absolute continuity and its a.e. derivative. Applying the uniform estimate to u_ε and then Young's inequality proves the norm bound. □

5.9.4 Continuity in a better spatial space (source Theorem 4)

Let U be bounded with smooth boundary and $m \ge 0$ an integer. If

$$u \in L^2(0,T;H^{m+2}(U)), \qquad u' \in L^2(0,T;H^m(U)),$$

then $u \in C([0,T];H^{m+1}(U))$, with the corresponding bound

$$\max_t \|u(t)\|_{H^{m+1}} \le C_{U,T,m}(\|u\|_{L^2(H^{m+2})} + \|u'\|_{L^2(H^m)}).$$

Expanded proof.

First $m = 0$. Choose the same compact-support extension operator E bounded both on L^2 and H^2, as constructed in §5.4. Set $v = Eu$. Bounded linear maps commute with Bochner integration, so testing in time gives $v' = Eu'$. For time-smooth v, integrate by parts in space on $\mathbb{R}^n$:

$$\frac{d}{dt}\|v\|_{H^1(\mathbb{R}^n)}^2 = 2(v - \Delta v, v')_{L^2}.$$

This is valid for H^2 functions by spatial approximation, with no boundary term on the whole space. Thus its absolute value is at most $C\|v\|_{H^2}\|v'\|_2$. Average an integrated identity in the initial time to obtain

$$\sup_t \|v(t)\|_{H^1}^2 \le C_T(\|v\|_{L^2(H^2)}^2 + \|v'\|_{L^2(L^2)}^2).$$

Reflect and convolve in time as in the preceding proof; now use continuity in L^2, which follows from $u \in W^{1,2}(0,T;L^2)$. The estimate on differences gives convergence in $C(H^1)$. Restrict to U and use the bounds for E.

能量计算的合法化

形式上把方程乘 u 会得到 $\frac{1}{2}(d/dt)\|u\|_2^2$。当 u_t 只有 H^{-1} 正则性时，必须先时间平滑、建立一致估计，再通过配对的 L^1 收敛取极限。

For general m, apply this result to every $D^\alpha u$, $|\alpha| \le m$. Spatial derivatives commute with the time weak derivative, as one checks against separated space-time test functions. Summing the finitely many H^1 estimates yields an equivalent H^{m+1} norm and the claimed continuity. □

空间与时间交换一阶

空间有 $m+2$ 阶、时间导数有 m 阶，得到时间连续的中间空间 H^{m+1}。空间分部积分把两个导数移到同一侧，正是这个阶数配平的来源。

5.10 Problems

Evans §5.10, Problems 1–21, printed pp. 289–292 (PDF pp. 306–309). Statements are concisely restated; solutions are editorial. Boundary smoothness is assumed where needed.

5.10.1 Problems 1–6: the basic language

1. Completeness of Hölder spaces. Prove $C^{k,\gamma}(\overline{U})$ is Banach.

Solution. The full proof in §5.1 identifies uniform limits of derivative components through coordinate-segment integration, then passes the Cauchy bound to the Hölder seminorm. Derivative compatibility and seminorm convergence are both needed, beyond uniform convergence of u_j. □

2. Hölder interpolation. For $0 < \beta < \gamma \le 1$, prove

$$\|u\|_{C^{0,\gamma}} \le \|u\|_{C^{0,\beta}}^{(1-\gamma)/(1-\beta)} \|u\|_{C^{0,1}}^{(\gamma-\beta)/(1-\beta)}.$$

Solution. The quotient factorization and two-term Hölder inequality in §5.1 give this formula. The weights sum to one and satisfy $\gamma = (1-\theta)\beta + \theta$ for $\theta = (\gamma-\beta)/(1-\beta)$. □

3. A pyramid on a square. On $U = (-1,1)^2$ the source's piecewise function is $u(x) = 1 - \max(|x_1|, |x_2|)$. Determine its $W^{1,p}$ membership.

Solution. On the four sectors where one signed coordinate is largest, Du is respectively $(-1,0), (1,0), (0,-1), (0,1)$. Integrate against a test function separately on the sectors. At each shared edge, values of u agree and outward normals are opposite, so the interface terms cancel. This bounded vector field is the weak gradient. Both u and Du are bounded on a finite-measure domain, so $u \in W^{1,p}(U)$ for all $1 \le p \le \infty$. □

4. One-dimensional Sobolev functions. Show $u \in W^{1,p}(0,1)$, $p < \infty$, has an absolutely continuous representative, and derive a Hölder estimate for $p > 1$.

Solution. Let g be its weak derivative and $G(x) = \int_0^x g(t)\,dt$. Fubini shows $G' = g$ weakly. Since $(u - G)' = 0$, Lemma 5.17 makes $u - G$ constant. The representative $c + G$ is absolutely continuous, with classical derivative g a.e. For every $x, y \in [0,1]$, Hölder gives

$$|u(x) - u(y)| \le \int_{\min(x,y)}^{\max(x,y)} |g| \le |x - y|^{1-1/p} \|g\|_{L^p(0,1)}.$$

For $p = 1$, absolute continuity persists but this argument supplies no fixed positive Hölder exponent. □

5. A smooth cutoff. For $V \Subset U$, construct $\zeta \in C_c^\infty(U)$ with $\zeta = 1$ on V.

Solution. Put $d = \min\{1, \operatorname{dist}(\overline{V}, \mathbb{R}^n \setminus U)\} > 0$ and $W = \{x : \operatorname{dist}(x, \overline{V}) < d/3\}$. For $\varepsilon < d/6$, $\zeta = \rho_\varepsilon * \mathbf{1}_W$ equals one on V, lies in $[0,1]$, and has compact support within distance $d/2$ of $\overline{V}$, hence inside U. □

6. A finite partition of unity. If $\overline{U}$ is compact and contained in $\bigcup_{i=1}^N V_i$, construct smooth ζ_i supported in V_i, with $0 \le \zeta_i \le 1$ and $\sum_i \zeta_i = 1$ on U.

Solution. Cover $\overline{U}$ by finitely many smaller balls with closures in the given cover. Choose nonnegative smooth bumps positive on these balls and group them by V_i, obtaining $\psi_i \in C_c^\infty(V_i)$. Their sum S is positive on a neighborhood O of $\overline{U}$. Choose $\chi \in C_c^\infty(O)$ equal to one near $\overline{U}$, with $0 \le \chi \le 1$. Set $\zeta_i = \chi\psi_i/S$ on O, zero elsewhere. The cutoff makes this smooth globally, and $\psi_i \le S$ gives $\zeta_i \le 1$. Their sum is χ, as required. □

5.10.2 Problems 7–10: boundary and interpolation estimates

7. Trace from a transverse vector field. If α is smooth and $\alpha \cdot \nu \ge 1$ on ∂U, prove the trace estimate for $1 \le p < \infty$.

分片求导还要查接口

每块导数有界仍不够，必须检查接口函数值相同。若有跳跃，弱导数就会多出沿接口集中的测度。

分母先保证为正

S 只在目标集邻域中确定为正；乘上 χ 后，在离开该邻域前商已为零，才可安全延拓。

Solution. For smooth u and $p > 1$, divergence and Young's inequality give

$$\begin{aligned}\int_{\partial U} |u|^p &\le \int_{\partial U} |u|^p \alpha \cdot \nu \\ &= \int_U |u|^p \operatorname{div} \alpha + p|u|^{p-2} u\, \alpha \cdot Du \\ &\le C \int_U (|u|^p + |Du|^p).\end{aligned}$$

The coefficients $\alpha, D\alpha$ are bounded on the compact closure. For $p = 1$, replace $|u|$ by $\sqrt{u^2 + \varepsilon^2}$; its gradient has magnitude at most $|Du|$. Let $\varepsilon \to 0$. Smooth boundary approximation gives the Sobolev estimate. □

8. No bounded trace on L^p alone. On a nonempty bounded C^1 domain, show no bounded linear map $L^p(U) \to L^p(\partial U)$, $p < \infty$, can agree with continuous boundary restriction.

Solution. Write $d(x) = \operatorname{dist}(x, \partial U)$. Choose continuous $\eta : [0, \infty) \to [0, 1]$ with $\eta(0) = 1$ and $\eta = 0$ on $[1, \infty)$. The functions $v_\delta(x) = \eta(d(x)/\delta)$ are continuous on $\overline{U}$, equal one on ∂U, and tend pointwise to zero in U. Dominated convergence gives $\|v_\delta\|_{L^p(U)} \to 0$. A bounded trace would force $0 < |\partial U|^{1/p} = \|T v_\delta\|_p \to 0$, a contradiction. □

边界薄层反例

函数支撑在越来越薄的边界层中，体积范数趋零，边界值却始终为 1。其梯度变大，说明迹估计为何需要导数信息。

9. An L^2 interpolation estimate. Prove $\|Du\|_2 \le C\|u\|_2^{1/2}\|D^2u\|_2^{1/2}$ for $C_c^\infty(U)$ and $H^2(U) \cap H_0^1(U)$.

Solution. For compactly supported smooth u,

$$\|Du\|_2^2 = -\int_U u\Delta u \le \|u\|_2 \|\Delta u\|_2 \le C_n \|u\|_2 \|D^2 u\|_2.$$

For $u \in H^2 \cap H_0^1$, the distributional identity $\int Du \cdot D\phi = -\int (\Delta u)\phi$ extends from tests to $\phi \in H_0^1$ by density, since both sides are continuous in that norm. Take $\phi = u$. No compactly supported H^2 approximation is needed. □

不能假设更强的逼近

紧支函数在 H^2 中的闭包还会限制法向导数，通常比 $H^2 \cap H_0^1$ 小。此处只在 H_0^1 中延伸测试恒等式，避免加上多余边界条件。

10. Nonlinear interpolation. For compactly supported smooth u, prove

$$\|Du\|_p \le C\|u\|_p^{1/2}\|D^2u\|_p^{1/2} \quad (2 \le p < \infty),$$

$$\|Du\|_{2p} \le C\|u\|_\infty^{1/2}\|D^2u\|_p^{1/2} \quad (1 \le p < \infty).$$

Solution. For $s \ge 2$, integrate by parts and differentiate $|Du|^{s-2}Du$ to get

$$\int |Du|^s \le C_s \int |u|\, |Du|^{s-2} |D^2 u|.$$

The vector map $z \mapsto |z|^{s-2}z$ is C^1 for $s \geq 2$, including the linear case $s = 2$. With $s = p > 2$, Hölder with exponents $p, p/(p-2), p$ gives $\|Du\|_p^p \leq C_p \|u\|_p \|Du\|_p^{p-2} \|D^2 u\|_p$. At $p = 2$ use Cauchy directly. Cancel the gradient factor unless zero, then take a square root. With $s = 2p$, use $\|u\|_\infty$ and exponents $p, p/(p-1)$ to get

$$\|Du\|_{2p}^{2p} \leq C_p \|u\|_\infty \|D^2 u\|_p \|Du\|_{2p}^{2p-2}.$$

For $p = 1$ the last factor is one. Cancellation gives the second result. □

5.10.3 Problems 11–16: the limits of the theory

11. Zero weak gradient. Prove on connected U that $Du = 0$ implies u constant a.e.

Solution. Lemma 5.17 gives the full argument: mollify on balls, identify constant limits, and propagate the common constant along a finite chain of overlapping balls. Connectedness enters the last step. □

12. Failure at $p = 1$. Show bounded L^1 difference quotients do not imply $W^{1,1}$ membership.

Solution. Take $U = (-2, 2)$, $V = (-1, 1)$, and $u = \mathbf{1}_{(0,2)}$ on U. For small h, $|D^h u| = 1/|h|$ on an interval of length $|h|$, and is zero elsewhere in V. Thus $\|D^h u\|_{L^1(V)} = 1$. But its distributional derivative is δ_0, so it is not in $W^{1,1}(V)$. □

13. Bounded gradient without global Lipschitz continuity.

Solution. Let U be the slit annulus $\{(r\cos\theta, r\sin\theta) : 1/2 < r < 1,\ 0 < \theta < 2\pi\}$ and put $u(r, \theta) = \theta$. This is smooth on U, with $|u| \leq 2\pi$ and $|Du| = 1/r \leq 2$. At $r = 3/4$, points of angles ε and $2\pi - \varepsilon$ approach each other, but their values differ by $2\pi - 2\varepsilon$. The function is in $W^{1,\infty}(U)$ and is not globally Lipschitz. The domain is connected but fails the regular boundary hypothesis at the slit. □

14. The source's unbounded critical example. For $n > 1$ verify $u(x) = \log\log(1 + 1/|x|) \in W^{1,n}(B_1)$.

Solution. For $r = |x|$,

$$|u'(r)| = \frac{1}{r(r+1)\log(1+1/r)} \asymp \frac{1}{r\log(1/r)} \quad (r \downarrow 0).$$

梯度控制路径长度

梯度界控制沿域内路径的变化；路径长度未必能由直线距离统一控制。裂缝反例不违背球内的局部 Lipschitz 结论。

The gradient integral near zero is comparable to $\int_0^{1/2} dr/(r\log(1/r)^n)$, finite for $n > 1$. The function integral has an exponentially decaying weight after $t = \log(1/r)$, and is also finite. Away from zero all quantities are bounded. The inner integration-by-parts term is bounded by $C\varepsilon^{n-1}|u(\varepsilon)| \to 0$, proving that the classical gradient off zero is its weak gradient. The function itself diverges as $r \downarrow 0$. □

15. A zero set of positive measure. If $u \in H^1(B_1)$ and $|\{u = 0\}| \geq \alpha > 0$, prove $\|u\|_2 \leq C_{n,\alpha}\|Du\|_2$.

Solution. Let $c = u_{B_1}$ and $Z = \{u = 0\}$. Poincaré gives

$$\alpha|c|^2 \leq \int_Z |u - c|^2 \leq \|u - c\|_2^2 \leq C_n\|Du\|_2^2.$$

Then $\|u\|_2 \leq \|u - c\|_2 + |B_1|^{1/2}|c|$ proves the estimate, with constant $C_n(1 + |B_1|^{1/2}\alpha^{-1/2})$. □

16. Whole-space Hardy inequality. For $n \geq 3$ prove

$$\int_{\mathbb{R}^n} \frac{|u|^2}{|x|^2} \leq \frac{4}{(n-2)^2}\int_{\mathbb{R}^n} |Du|^2, \qquad u \in H^1(\mathbb{R}^n).$$

Solution. First take compactly supported smooth u. Integrate the identity

$$\operatorname{div}\left(\frac{x}{|x|^2}\right) = \frac{n-2}{|x|^2}$$

outside B_ε. The inner boundary term is $O(\varepsilon^{n-2})$ and vanishes; there is no outer term. Thus

$$(n-2)\int \frac{u^2}{|x|^2} = -2\int u\,Du \cdot \frac{x}{|x|^2} \leq 2\left(\int \frac{u^2}{|x|^2}\right)^{1/2}\|Du\|_2.$$

Cancel and square. For general u, compactly supported smooth H^1 approximations are available by cutoff and convolution. The estimate on differences makes $u_j/|x|$ Cauchy in L^2; an a.e. convergent subsequence identifies its limit with $u/|x|$. □

5.10.4 Problems 17–19: nonlinear operations

17. Chain rule. On bounded U, let $F \in C^1(\mathbb{R})$ have bounded derivative. Prove $F(u) \in W^{1,p}$ and $D(F(u)) = F'(u)Du$.

Solution. The bound $|F(s)| \le |F(0)| + \|F'\|_\infty |s|$ gives integrability. For finite p, smooth interior approximations converge in local $W^{1,p}$ and, along a subsequence, a.e. The Lipschitz bound gives $F(u_j) \to F(u)$ in local L^p. Split the gradients:

$$F'(u_j)Du_j - F'(u)Du = F'(u_j)(Du_j - Du) + (F'(u_j) - F'(u))Du.$$

The first term tends to zero by boundedness of F'; the second by dominated convergence with majorant $2\|F'\|_\infty |Du|$. Pass to the weak identity locally. The candidate derivative is globally L^p, so this proves membership on U. For $p = \infty$, use finite local exponents and retain the resulting L^∞ bounds on function and gradient. □

用常数检验边界项

球域 Hardy 需要低阶项，因为常数在球上属于 H^1；全空间非零常数不在 L^2，所以不反驳纯梯度型估计。

18. Positive parts and level sets. Prove $|u|, u^+, u^- \in W^{1,p}(U)$ and

$$Du^+ = \mathbf{1}_{\{u>0\}}Du, \qquad Du^- = -\mathbf{1}_{\{u<0\}}Du, \qquad Du = 0 \text{ a.e. on } \{u = 0\}.$$

Solution. The function

$$F_\varepsilon(s) = \begin{cases} \sqrt{s^2 + \varepsilon^2} - \varepsilon, & s \ge 0, \\ 0, & s < 0 \end{cases}$$

is C^1, with $|F_\varepsilon'| \le 1$, $F_\varepsilon' \to \mathbf{1}_{\{s>0\}}$ pointwise, and $|F_\varepsilon(s) - s^+| \le \varepsilon$. Problem 17 and dominated convergence show for finite p that

$$F_\varepsilon(u) \to u^+ \text{ in } L^p, \qquad D(F_\varepsilon(u)) \to \mathbf{1}_{\{u>0\}}Du \text{ in } L^p.$$

The weak identity yields the positive-part formula. Apply it to $-u$ for the negative part. Differentiate $u = u^+ - u^-$ to obtain $Du = \mathbf{1}_{\{u\neq 0\}}Du$, proving the level-set assertion. Use $|u| = u^+ + u^-$ for the absolute value. For $p = \infty$ use a finite exponent on bounded U and the derivative bounds to recover $W^{1,\infty}$ membership. □

链式法则如何取极限

$F'(u_j) \to F'(u)$ 只是逐点信息；第二项需要由 $2\|F'\|_\infty |Du|$ 支配，第一项则依赖梯度的强收敛。

19. A second level-set argument for H^1. Let ϕ be smooth, bounded, nondecreasing, with bounded derivative, and $\phi(s) = s$ for $|s| \le 1$. Set $u_\varepsilon = \varepsilon\phi(u/\varepsilon)$.

Solution. Since $\phi(0) = 0$ and ϕ' is bounded, $|u_\varepsilon| \le \|\phi'\|_\infty |u|$ while $|u_\varepsilon| \le \varepsilon\|\phi\|_\infty \to 0$ pointwise. Dominated convergence gives $u_\varepsilon \to 0$ in L^2, even when U has infinite measure. Also $Du_\varepsilon = \phi'(u/\varepsilon)Du$ is bounded in L^2. Every sequence $\varepsilon_j \to 0$ has a weak H^1 subsequence by reflexivity; its limit is zero by strong L^2 convergence. Thus the whole family converges weakly to zero. Pairing its gradient with Du gives

$$0 \le \int_{\{u=0\}} |Du|^2 \le \int_U \phi'(u/\varepsilon)|Du|^2 = \int_U Du_\varepsilon \cdot Du \to 0.$$

We used $\phi' \ge 0$ and $\phi'(0) = 1$, without needing any pointwise limit of $\phi'(u/\varepsilon)$ away from $\{u = 0\}$. □

不能漏掉零水平集

$\{u = 0\}$ 可能有正测度，不能只在正负区域求导就结束。先证明正负部公式，再由 $u = u^+ - u^-$ 推出该集合上梯度为零。

5.10.5 Problems 20–21: Fourier estimates

20. Boundedness above half the dimension. Show $H^s(\mathbb{R}^n) \hookrightarrow L^\infty$ for $s > n/2$.

Solution. Put $w_s(\xi) = (1 + |\xi|^2)^{s/2}$. Cauchy–Schwarz gives

$$\|\hat{u}\|_1 \le \|w_s\hat{u}\|_2 \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s}\, d\xi \right)^{1/2} \le C_{n,s}\|u\|_{H^s}.$$

The radial integral at infinity is $\int_1^\infty r^{n-1-2s}\, dr$, finite exactly when $2s > n$. The inverse Fourier integral is bounded and continuous, with supremum at most $(2\pi)^{-n/2}\|\hat{u}\|_1$, and agrees with u in L^2 by Plancherel/inversion. □

半维数的来源

要让逆变换绝对可积，频率权重的倒数需属于 L^2；径向积分恰好给出条件 $2s > n$，而不是随意选的正则性门槛。

21. The algebra property. For $s > n/2$, prove $\|uv\|_{H^s} \le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}$.

Solution. For Schwartz functions the triangle inequality gives $w_s(\xi) \le C_s(w_s(\eta) + w_s(\xi - \eta))$. The product-convolution formula implies

$$w_s|\widehat{uv}| \le C_{n,s}\big[(w_s|\hat{u}|) * |\hat{v}| + |\hat{u}| * (w_s|\hat{v}|)\big].$$

Young's $L^2 * L^1 \to L^2$ inequality and Problem 20 give

$$\begin{aligned} \|uv\|_{H^s} &\le C_{n,s}\big(\|u\|_{H^s}\|\hat{v}\|_1 + \|\hat{u}\|_1\|v\|_{H^s}\big) \\ &\le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}. \end{aligned}$$

For general u, v, approximate their transforms by smooth compactly supported functions in weighted L^2. Their inverse transforms are Schwartz. The bilinear estimate makes their products Cauchy in H^s. Problem 20 gives uniform convergence of the factors, and

$$\|u_j v_j - uv\|_2 \le \|u_j\|_\infty \|v_j - v\|_2 + \|u_j - u\|_\infty \|v\|_2 \to 0.$$

Thus the H^s limit is the actual product uv. □

乘积为何不丢阶

$s > n/2$ 提供频率的 L^1 控制，使卷积可用 $L^2 * L^1$ 估计。因此多项式非线性可以在同一个 H^s 空间内处理。

5.11 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 5, printed pp. 239–292 (local English PDF pp. 256–309). The chapter order and source theorem identifiers follow that edition. The Chinese scan has Chapter 5 at printed pp. 169–207 (PDF pp. 187–225); the chapter opening and its list of section starts were visually checked. It is a terminology reference, not the source of the English prose written here.

For Sobolev theory, Evans's §5.11 points to Gilbarg–Trudinger (Chapter 7), Lieb–Loss, Ziemer, and Evans–Gariepy. For BMO the source points to Stein, Chapter IV; for time-dependent spaces to Temam, pp. 248–273; and for whole-space Hardy to Tartar, Chapter 17. These are source pointers, not claims that those books have been independently consulted for this draft.

Proof dependencies and review boundary. This chapter invokes the measure-theoretic and functional-analytic prerequisites listed at its opening, including Lebesgue differentiation, Plancherel, Riesz representation, reflexivity, and Arzel–Ascoli. These foundational theorems are explicitly invoked rather than reproved. Other displayed chapter theorems have reconstructed proofs here. All 21 exercises have editorial solutions or exact pointers to full arguments earlier in the chapter. Source comparison, mathematical self-checking, compilation, and visual inspection have separate scopes. This study draft has not received independent mathematical review.